KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME: **WORKED SOLUTIONS**

TEACHER:

2026

YEAR 12
HSC TASK 1

Mathematics Standard 2

Ratios and rates • Trigonometry • **WORKED SOLUTIONS – TEACHER COPY**

General Instructions

- Working time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks: 45

Section I – 10 marks (pages 3–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 35 marks (pages 7–13)

- Attempt Questions 11–23
- Allow about 45 minutes for this section

Outcomes assessed

MST-12-S2-04 applies trigonometry to solve problems involving right-angled and non-right-angled triangles
MST-12-S2-05 solves practical problems involving ratios and rates
MAO-WM-01 develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

Section I – Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:	2 + 4 =	(A) 2	(B) 6	(C) 8	(D) 9
		A ○	B ●	C ○	D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

		A ●	B ●	C ○	D ○

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

		A ●	B ● ↙ correct	C ○	D ○

1.	A ○	B ●	C ○	D ○
2.	A ○	B ○	C ●	D ○
3.	A ○	B ○	C ○	D ●
4.	A ●	B ○	C ○	D ○
5.	A ○	B ●	C ○	D ○
6.	A ○	B ○	C ○	D ●
7.	A ○	B ○	C ●	D ○
8.	A ●	B ○	C ○	D ○
9.	A ○	B ○	C ○	D ●
10.	A ○	B ○	C ●	D ○

Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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1 What is the ratio 1.2 L : 400 mL in simplest form?

A. 1 : 3

B. 3 : 1

C. 3 : 10

D. 30 : 1

1.2 L = 1200 mL, so 1200 : 400 = 3 : 1 **Answer: B**

2 On a map, a distance of 4 cm represents an actual distance of 2 km. What is the scale of the map?

A. 1 : 500

B. 1 : 5000

C. 1 : 50 000

D. 1 : 500 000

4 cm : 2 km = 4 cm : 200 000 cm = 1 : 50 000 **Answer: C**

3 What is the compass bearing S35°W written as a true bearing?

A. 035°

B. 145°

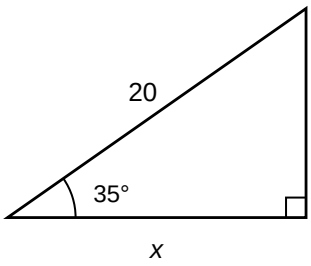
C. 235°

D. 215°

S35°W → 180° + 35° = 215° **Answer: D**

4 Which expression gives the value of x in the triangle shown?

NOT TO SCALE



A. 20 cos 35°

B. 20 sin 35°

C. 20 ÷ cos 35°

D. 20 tan 35°

x is adjacent to 35° and 20 is the hypotenuse: cos 35° = x/20, so x = 20 cos 35° **Answer: A**

5 Olive oil is sold in four bottle sizes. Which bottle is the best value for money?



A. 500 mL for $6.80 B. 750 mL for $9.90 C. 1 L for $13.50 D. 2 L for $26.80

Per litre: A $13.60, B $13.20, C $13.50, D $13.40 → B is cheapest per litre **Answer: B**

6 Fisheries officers catch, tag and release 60 bream in Cudgen Creek. A week later they catch 50 bream, and 6 of these are tagged. What is the best estimate of the number of bream in the creek?

A. 110

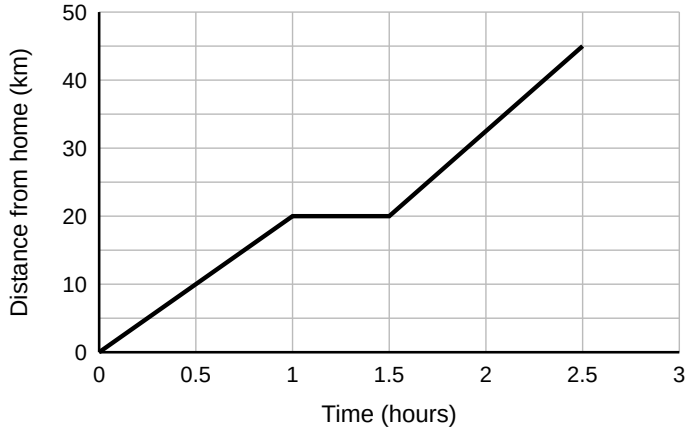
B. 300

C. 360

D. 500

6/50 = 60/N, so N = 60 × 50 ÷ 6 = 500 **Answer: D**

7 The distance–time graph shows Mia's bike ride from home to Hastings Point.



What was Mia's speed during the final stage of her ride?

A. 18 km/h

B. 20 km/h

C. 25 km/h

D. 45 km/h

Final stage: (45 − 20) km ÷ (2.5 − 1.5) h = 25 km/h **Answer: C**

8 A 2400 W heater is used for 3 hours. Electricity costs 30 c/kWh. What does it cost to run the heater for this time?

A. $2.16

B. $7.20

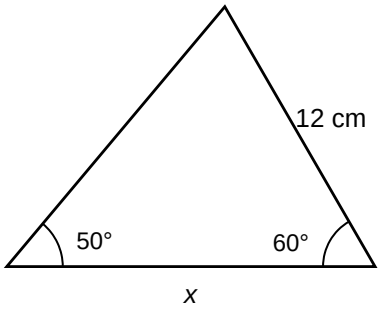
C. $21.60

D. $216.00

2400 W = 2.4 kW; 2.4 × 3 = 7.2 kWh; 7.2 × $0.30 = $2.16 **Answer: A**

9 What is the value of x, correct to one decimal place?

NOT TO SCALE



A. 9.8 cm

B. 10.6 cm

C. 13.6 cm

D. 14.7 cm

Third angle = 180° − 50° − 60° = 70°; x = 12 sin 70° ÷ sin 50° = 14.72… **Answer: D**

10 A triangle has an area of 30 m^2. Two of its sides are 8 m and 10 m, and the angle between these two sides is acute. What is the size of this angle, correct to the nearest degree?

A. 22°

B. 41°

C. 49°

D. 131°

½ × 8 × 10 × sin θ = 30 → sin θ = 0.75 → θ = 48.59…° ≈ 49° (acute) **Answer: C**

Section II

35 marks
Attempt Questions 11–23
Allow about 45 minutes for this section

Answer each question in the spaces provided. Your responses should include relevant mathematical reasoning and/or calculations.

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11 Ava, Ben and Cal buy a raffle ticket together. Ava pays \$3, Ben pays \$5 and Cal pays \$7. The ticket wins a prize of \$1200, which they share in the same ratio as the amounts they paid. How much does Cal receive? **2**

Total parts = 3 + 5 + 7 = 15

One part = \$1200 ÷ 15 = \$80

Cal = 7 × \$80 = \$560

12

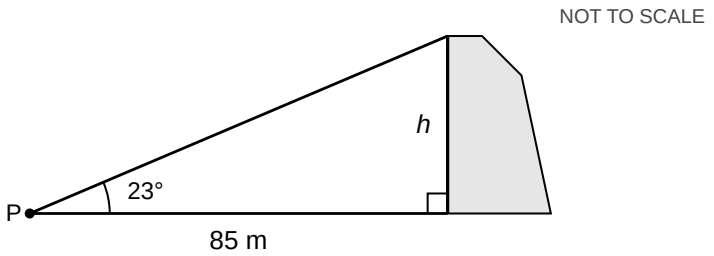
a. Write the true bearing 290° as a compass bearing. **1**

360° − 290° = 70°, so 290° = N70°W

b. The bearing of Cook Island from Fingal Head is 052°. What is the bearing of Fingal Head from Cook Island? **1**

052° + 180° = 232°

13 From a point P on the sand, 85 m from the base of a headland, the angle of elevation of the top of the headland is 23°. Find the height, *h*, of the headland, correct to the nearest metre. **2**



tan 23° = h/85

h = 85 × tan 23° = 36.080…

h ≈ 36 m

14 Jordan drives from Kingscliff to Sydney, a distance of approximately 820 km. Jordan's car uses 7.4 L/100 km.

a. How many litres of fuel will Jordan's car use for this trip? **1**

Fuel = 820 ÷ 100 × 7.4

= 60.68 L

b. Fuel costs $2.06 per litre. Jordan says the fuel for the trip will cost less than $120. Is Jordan correct? Justify your answer with a calculation. **1**

Cost = 60.68 × $2.06 = $125.00

$125.00 > $120, so Jordan is not correct.

15 A 2500 W air conditioner runs for 6 hours a day for the 90 days of summer. Electricity costs 32 c/kWh. Find the cost of running the air conditioner for the summer. **2**

2500 W = 2.5 kW; energy = 2.5 kW × 6 h × 90 days = 1350 kWh

Cost = 1350 × $0.32

= $432

16 On a map with a scale of 1 : 25 000, the coastal walking track from Kingscliff to Cudgen Headland measures 14.6 cm.

a. Find the actual length of the track in kilometres. **1**

Actual = 14.6 cm × 25 000 = 365 000 cm

= 3650 m = 3.65 km

b. Ella walks the track at an average speed of 4.5 km/h. How long does the walk take, correct to the nearest minute? **2**

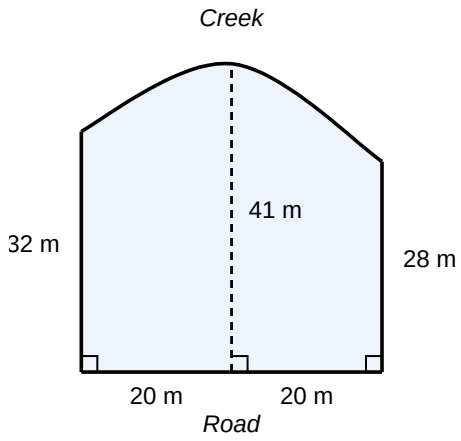
T = D ÷ S = 3.65 ÷ 4.5 = 0.8111… h

0.8111… × 60 = 48.67 minutes

≈ 49 minutes

17 A block of land lies between a straight road and a creek. Measurements taken from the road are shown in the diagram. Use two applications of the trapezoidal rule to estimate the area of the block. **2**

NOT TO SCALE



$A \approx 20/2 \times (32 + 41) + 20/2 \times (41 + 28)$

$= 10 \times 73 + 10 \times 69 = 730 + 690$

$= 1420 \text{ m}^2$

18 A garden shed has a flat roof that is 8 m long and 5.5 m wide. During a storm, 24 mm of rain falls on the roof, and 90% of this rain is collected in a tank. How many litres of water are collected? **2**

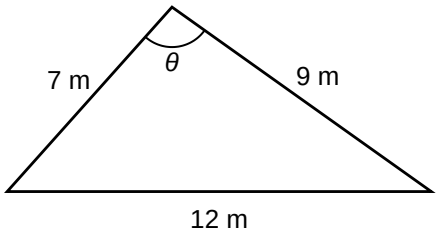
Roof area $= 8 \times 5.5 = 44 \text{ m}^2$; 24 mm = 0.024 m

$V = Ah = 44 \times 0.024 = 1.056 \text{ m}^3$; 90%: $0.9 \times 1.056 = 0.9504 \text{ m}^3$

= 950.4 L ($1 \text{ m}^3 = 1000$ L)

19 Find the size of angle θ in the triangle below, correct to the nearest degree. **2**

NOT TO SCALE



$\cos\theta = (7^2 + 9^2 - 12^2) \div (2 \times 7 \times 9)$

$= -14 \div 126 = -0.1111\ldots$

$\theta = 96.379\ldots° \approx 96°$

20 A 6 L container is full of a drink made from cordial and water mixed in the ratio 1 : 5. Then 1.5 L of the drink is poured out, and the container is filled back up to 6 L with water only. What is the new ratio of cordial to water, in simplest form? **3**

Drink left = 6 − 1.5 = 4.5 L

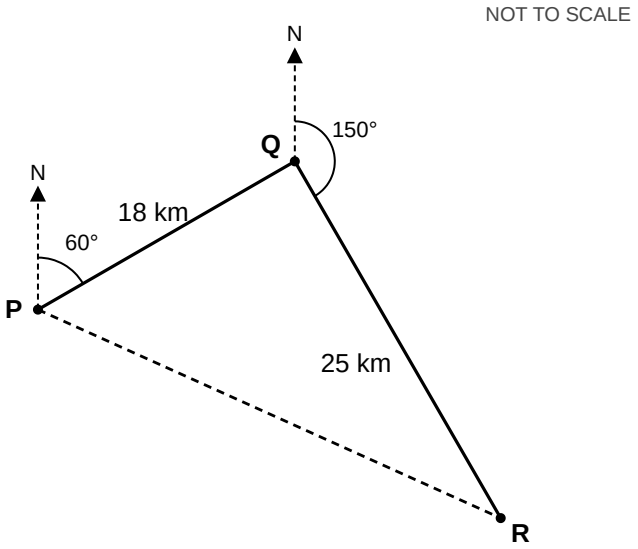
Cordial = 1/6 × 4.5 = 0.75 L; water = 5/6 × 4.5 = 3.75 L

Add 1.5 L of water: water = 3.75 + 1.5 = 5.25 L

Cordial : water = 0.75 : 5.25

= 1 : 7

21 A boat sails 18 km from port P to Q on a bearing of 060°. It then sails 25 km from Q to R on a bearing of 150°.



a. Show that ∠PQR = 90°. **1**

Bearing of P from Q = 060° + 180° = 240°

∠PQR = 240° − 150° = 90°

b. Find the distance PR, correct to one decimal place. **1**

$PR^2 = 18^2 + 25^2 = 949$

PR = √949 = 30.805… ≈ 30.8 km

c. Find the bearing of R from P, correct to the nearest degree. **2**

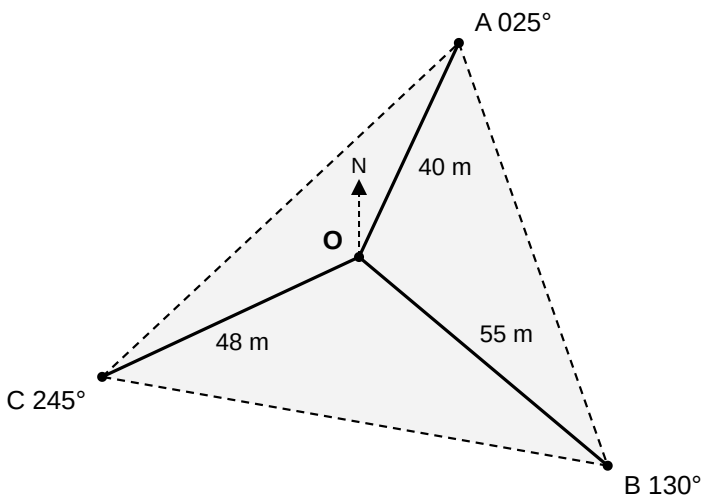
tan ∠QPR = 25/18, so ∠QPR = 54.246…°

Bearing of R from P = 060° + 54.246…°

= 114.246…° ≈ 114°

22 A compass radial survey of a triangular field ABC is shown. The bearing and distance of each corner from the point O are given.

NOT TO SCALE



a. Show that the size of ∠BOC is 115°. **1**

∠BOC = 245° − 130° = 115°

b. Find the area of triangle AOB, correct to the nearest square metre. **2**

∠AOB = 130° − 25° = 105°

A = ½ × 40 × 55 × sin 105°

= 1062.518… ≈ 1063 m²

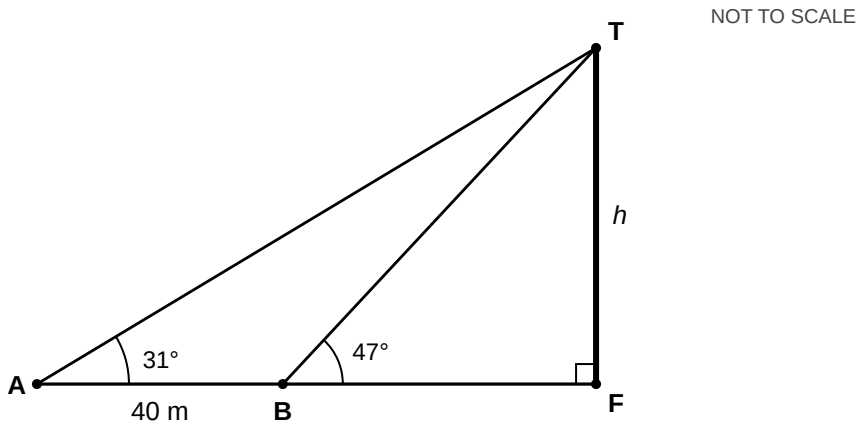
c. A fence is to be built along the boundary AB. Find the length of the fence, correct to one decimal place. **2**

$AB^2 = 40^2 + 55^2 - 2 \times 40 \times 55 \times \cos 105°$

= 5763.80…

AB = √5763.80… = 75.919… ≈ 75.9 m

23 From point A on level ground, the angle of elevation of the top, T, of a telecommunications tower is 31°. From point B, 40 m closer to the base F of the tower, the angle of elevation of T is 47°.



a. Show that ∠ATB = 16°. **1**

∠ABT = 180° − 47° = 133° (straight angle); ∠ATB = 180° − 31° − 133° = 16° (angle sum)

b. Use the sine rule in triangle ABT to find the length of BT, correct to one decimal place. **2**

BT ÷ sin 31° = 40 ÷ sin 16°

BT = 40 × sin 31° ÷ sin 16°

= 74.741… ≈ 74.7 m

c. Hence find the height, *h*, of the tower, correct to the nearest metre. **1**

sin 47° = h ÷ BT, so h = 74.741… × sin 47°

= 54.66… ≈ 55 m

End of Exam

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

NSW Education Standards Authority

2023 **HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics Standard 1
Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

Absolute error = $\frac{1}{2} \times$ precision

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}\left(a + b\right)$$

$$A \approx \frac{h}{2}\left(d_f + d_l\right)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV\left(1 + r\right)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0\left(1 - r\right)^n$$

Statistical Analysis

An outlier is a score

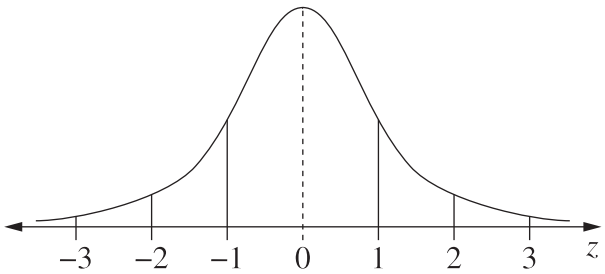
less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3